

A thermal LBM-LES model in body-fitted coordinates: Flow and heat transfer around a circular cylinder in a wide Reynolds number range

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ABSTRACT

In order to apply the lattice Boltzmann method (LBM) for modeling passive heat transfer at high Reynolds numbers, a number of models were proposed by introducing the large eddy simulation (LES) into the LBM framework to improve numerical stability. Our study finds that the generalized form of interpolation-supplemented LBM (GILBM), likewise, can locally modify the dimensionless relaxation time, thus enhancing the numerical stability at high Reynolds numbers. Given additional advantages of the GILBM in dealing with complicated geometries and improving computing accuracy, a thermal LBM-LES model in body-fitted coordinates is established in this paper. Numerical validation is performed by investigating the flow and heat transfer around a circular cylinder over a wide range of Reynolds numbers. The obtained results agree well with both experimental and numerical data in the previous work. Meanwhile, the effects of Reynolds number and Prandtl number on thermodynamic features of flow past a circular cylinder are revealed. It is found out that when the Reynolds number exceeds the critical value, the local Nusselt number fluctuates rapidly in a specific region of the rear cylinder surface affected by the Prandtl number. In the near-wake region, the temperature field exhibits significant dependence on the Prandtl number at moderate Reynolds numbers, while such effects turn to be slight at high Reynolds numbers.

1. Introduction

As an alternative numerical scheme to traditional CFD approaches, the lattice Boltzmann method (LBM), based on the mesoscopic kinetic equations, has been established to be an effective and promising numerical tool for simulating single/multi-phase fluid flow and heat transfer in the past few decades (Chen and Doolen, 2003; Higuera and Jiménez, 1989; Li et al., 2014a; Ma et al., 2018; Zhou et al., 2019). Meanwhile, distinct advantages of the LBM, including explicit scheme, parallel computing, and simple boundary treatment, attract many researchers to use and develop it for more practical applications. Generally, the LBM can be used to solve flow and heat transfer issues at relative low Reynolds numbers. The using of the LBM for thermal flow at high Reynolds numbers, without modeling unresolved small-scale effects on large-scale dynamics, results in numerical instability (Hou et al., 1996). However, there is a great demand to solve the turbulent thermal flow issues, which are frequently encountered in the nature and engineering fields. Hence, the improvement of the LBM for fluid flow and heat transfer at high Reynolds numbers has generated considerable research interest.

There are three typical methods available for turbulent flow simulation, including the direct numerical simulation (DNS), the Reynolds average numerical simulation (RANS), and the large eddy simulation (LES). Amongst these methods, the LES proposed first by Deardorff (1973) is the easiest one that can be introduced into the LBM framework. Initial attempt was done in the work of Hou et al. (1996). The well-known standard Smagorinsky model (Smagorinsky, 1963) was directly applied to the LBM by filtering the particle distribution function with a standard box filter. To account for the nonlinear term in the equilibrium distribution, the total dimensionless relaxation time determined by the physical and eddy viscosities is adopted in the evolution equation. Unlike the work of Hou et al., Chen (2009) proposed a novel LES-based LB model in which the target macroscopic equations were replaced by the vorticity-stream function equations. Due to the essential features of vorticity-stream function equations, this model has good numerical efficiency and stability in turbulent flow simulation. Moreover, Premnath et al. (2013) evaluated a LB model based on a multiblock approach for efficient large eddy simulation of 3D external flow past a circular cylinder with Reynolds number up to 3900. The

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obtained statistics of the mean flow and turbulent fluctuations were all in good agreement with prior data. It is worth noting that the recently developed Discrete Unified Gas-Kinetic Scheme (DUGKS), which shares the common kinetic origin with the LBM, also shows good stability in studying turbulent flow (Wang et al., 2016). Benefiting from its finite volume method (FVM) nature, non-uniform meshes can be easily employed in the DUGKS without loss of accuracy (Wang et al., 2015).

As for applications of the LBM in modeling turbulent thermal flow, most of the previous work was limited in the topic of natural convection in a closed cavity. For example, Horvat et al. (2001) proposed a thermal LBGK model based on the LES to simulate turbulent natural convection due to internal heat generation in a square cavity. The corresponding results indicated good numerical stability for the thermal flow simulation at high Rayleigh numbers. Chen et al. (2012) developed a new LES-based LB model to simulate turbulent double-diffusive convection in a square cavity. The power-law relationship among the Nusselt number, the ratio of buoyancy force, and the Rayleigh number is concluded in their work. In terms of the turbulent passive heat transfer, Chang et al. (2013) introduced the LES into the LBM to simulate a closed lid-driven flow and a backward-facing step thermal flow in both laminar and turbulent flow regimes. The temperature field is coupled with the velocity field by defining the sub-grid Prandtl Number. Furthermore, Wu et al. (2011) applied a LES-MRT thermal LB model to simulate passive heat transfer in a turbulent channel flow, which demonstrates great reliability in studying turbulence. A successful modeling of turbulent channel flow and heat transfer was also reported by Ren et al. (2018). In their work, the dynamic Vreman model was incorporated under the LBM framework for the first time.

According to the work of Sterling and Chen (1996), the dimensionless relaxation time must be greater than 0.5 corresponding to a positive physical viscosity, and the numerical stability is weakened once it is close to 0.5. It is known that the dimensionless relaxation time approaches 0.5 with the increasing value of Reynolds number. Namely, high Reynolds numbers could cause numerical instability. To remove such negative effects stemmed from high Reynolds numbers, the dimensionless relaxation time is locally increased by introducing the eddy viscosity in the above-mentioned LES-based models. Notably, the generalized form of interpolation-supplemented LBM (GILBM) proposed by Imamura et al. (2005a, b) presents similar modification of the dimensionless relaxation time. In this method, the time step δt is locally determined, and previous study indicates that this local time step δt is generally less than one ($\delta t < 1.0$), which is different from that in the standard LB model ($\delta t = 1.0$) (Li et al., 2015). So then, the dimensionless relaxation time τ is increased as it is related to the physical viscosity ν and the local time step δt as $\tau = \nu/(c_s^2 \delta t) + 0.5$. Meanwhile, Li et al. (2014b) thoroughly assessed the GILBM by studying the fluid flow in irregular domains. It was found out that as the Reynolds number increases, both the accuracy and efficiency of the GILBM have a better performance than those of the standard LBM. Additionally, the GILBM shows distinct advantages in dealing with thermal flow occurring in domain of complicated boundaries, and the local grid refinement techniques can be easily involved in it (Filippova and Hänel, 2000; Yu et al., 2002).

Based on the foregoing analysis, it is necessary to combine the LES with the GILBM in consideration of the positive effects on numerical stability and efficiency at high Reynolds numbers. Unfortunately, to the best of our knowledge, few studies have been reported in this aspect. Therefore, the aim of this paper is to develop a thermal LBM-LES model in body-fitted coordinate system by introducing the LES into the GILBM framework. To reach this goal, the improved model should be validated by a well-documented test case. Compared with various cases, the flow and heat transfer around a circular cylinder is more appropriate and challenging. Especially at high Reynolds numbers, the flow past a circular cylinder involves complex flow features, such as transition and large-scale vortex motion in the wake. Besides, detailed experimental and numerical data can be referred in the archival literature

(Cantwell and Coles, 1983; Breuer, 2000; Patnana et al., 2010; Catalano et al., 2003). Therefore, in the current work, a thorough numerical investigation of the flow and heat transfer around a circular cylinder over a wide range of Reynolds numbers is conducted to test the performance of new model. Meanwhile, the influences of Reynolds number and Prandtl number on thermodynamic features of the flow past a surface-heated circular cylinder are analyzed, respectively. The rest of the article is organized as follows. In Section 2, a thermal LBM-LES model in body-fitted coordinate system is derived. Subsequently, numerical simulation is performed in Section 3. Finally, some conclusions are summarized in Section 4.

2. Methodology

2.1. GILBM-based thermal LB model

In this section, a thermal LB model, based on the GILBM for studying fluid flow and heat transfer in irregular geometries, is derived. According to the idea of the GILBM, the evolution equations in the physical domain ($\mathbf{x} = (x_1, x_2, x_3) = (x, y, z)$) should be transformed to the computational domain ($\xi = (\xi_1, \xi_2, \xi_3) = (\xi, \eta, \gamma)$). For simplicity, only the evolution equations in the computational domain are presented here. The continuity and momentum equations and the energy equation are recovered by two sets of particle distribution functions f_i and h_i , respectively. First, the corresponding collision equations are expressed as

$$f_i^*(\xi, t) = f_i(\xi, t) - \frac{1}{\tau_f} [f_i(\xi, t) - f_i^{eq}(\xi, t)] \quad (1)$$

$$h_i^*(\xi, t) = h_i(\xi, t) - \frac{1}{\tau_T} [h_i(\xi, t) - h_i^{eq}(\xi, t)] \quad (2)$$

where $f_i^*(\xi, t)$ and $h_i^*(\xi, t)$ are the post-collision particle distribution functions at position ξ and time t in the computational domain. τ_f and τ_T are dimensionless relaxation times related to f_i and h_i , respectively. For the D3Q19 model, the equilibrium distribution functions f_i^{eq} and h_i^{eq} are calculated as

$$f_i^{eq} = \rho \omega_i \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (3)$$

$$h_i^{eq} = T \omega_i \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right] \quad (4)$$

where ω_i is the weighting coefficient, $\mathbf{e}_i$ stands for the discrete velocity of i th direction (Qian et al., 1992). $c_s^2 = c^2/3$ is the lattice velocity of sound. ρ , T , and $\mathbf{u}$ are macroscopic density, temperature, and velocity, respectively.

Fig. 1 shows the distribution of the discrete velocity vector $\mathbf{e}_7$ in both the physical and computational domains. It is observed that the discrete velocity vectors are constant in the physical domain, while they vary with the lattice position in the computational domain (Imamura et al., 2005a). Hence, the contravariant velocity $\tilde{\mathbf{e}}_i$ is defined to replace the standard discrete velocity $\mathbf{e}_i$ as

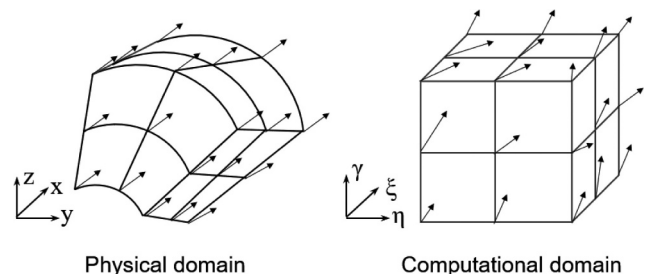


Fig. 1. Discrete velocity vectors on physical and computational domains.

$$\tilde{e}_{i,\alpha} = e_{i,\beta} \frac{\partial \xi_\alpha}{\partial x_\beta} \quad (5)$$

The derivatives in Eq. (5) can be obtained through a modified Jacobian conversion as

$$\frac{\partial \xi_\alpha}{\partial x_\beta} = \frac{A_{\alpha\beta}}{J} \quad (6)$$

where J is the Jacobian determinant, $A_{\alpha\beta}$ is a transformation matrix, and they are defined as

$$J = \begin{vmatrix} x_\xi & y_\xi & z_\xi \\ x_\eta & y_\eta & z_\eta \\ x_\gamma & y_\gamma & z_\gamma \end{vmatrix}, \quad A_{\alpha\beta} = \begin{bmatrix} y_\eta z_\gamma - y_\gamma z_\eta & x_\gamma z_\eta - x_\eta z_\gamma & x_\eta y_\gamma - x_\gamma y_\eta \\ y_\gamma z_\xi - y_\xi z_\gamma & x_\xi z_\gamma - x_\gamma z_\xi & x_\gamma y_\xi - x_\xi y_\gamma \\ y_\xi z_\eta - y_\eta z_\xi & x_\eta z_\xi - x_\xi z_\eta & x_\xi y_\eta - x_\eta y_\xi \end{bmatrix} \quad (7)$$

The second-order central-difference scheme is adopted to calculate the derivatives in Eqs. (5)–(7) for maintaining the accuracy in space. Subsequently, the particle streaming distance $\Delta \xi_{up,i}$ is calculated by integrating the contravariant velocity over a time step δt as

$$\Delta \xi_{up,i} = \int_0^{\delta t} d\xi_i = \int_0^{\delta t} \tilde{e}_i dt \quad (8)$$

The above integral is achieved by the two-step Runge-Kutta method (Imamura et al., 2005a) as

$$\text{1st-step: } \Delta \xi_{up,i}^{(1)} = \frac{1}{2} \delta t \tilde{e}_i(\xi) \quad (9a)$$

$$\text{2nd-step: } \Delta \xi_{up,i} = \delta t \tilde{e}_i \left(\xi - \Delta \xi_{up,i}^{(1)} \right) + O(\delta t^3) \quad (9b)$$

Then, the streaming equations become

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) \quad (10)$$

$$h_i(\xi, t + \delta t) = h_i^*(\xi - \Delta \xi_{up,i}, t) \quad (11)$$

In Eqs. (9–11), generally, the positions $\xi - \Delta \xi_{up,i}$ and $\xi - \Delta \xi_{up,i}^{(1)}$ depending on the contravariant velocity are not located at the lattice points in the computational domain, so an interpolation procedure is extremely needed. As suggested by Imamura et al. (2005a), the second-order upwind quadratic interpolation is adopted to suppress the numerical dissipation. The interpolation functions are given as

$$f_i(\xi, t + \delta t) = \sum_{k=0}^2 \sum_{l=0}^2 \sum_{s=0}^2 a_{i,k,\xi} a_{i,l,\eta} a_{i,s,\gamma} f_i^*(\xi_{m+k-md}, \eta_{n+l-nd}, \xi_{p+s-pd}, t) \quad (12)$$

$$h_i(\xi, t + \delta t) = \sum_{k=0}^2 \sum_{l=0}^2 \sum_{s=0}^2 a_{i,k,\xi} a_{i,l,\eta} a_{i,s,\gamma} h_i^*(\xi_{m+k-md}, \eta_{n+l-nd}, \xi_{p+s-pd}, t) \quad (13)$$

where $md = \text{sgn}(\Delta \xi_{up,i,\xi})$, $nd = \text{sgn}(\Delta \xi_{up,i,\eta})$ and $pd = \text{sgn}(\Delta \xi_{up,i,\gamma})$ are the coefficients to determine the lattice points for interpolation. As for the internal lattice nodes, the interpolation coefficients $a_{i,m,\xi}$, $a_{i,l,\eta}$ and $a_{i,s,\gamma}$ are given as

$$\begin{aligned} a_{i,0,\alpha} &= \frac{1}{2} (|\Delta \xi_{up,i,\alpha}| - 1) (|\Delta \xi_{up,i,\alpha}| - 2) \\ a_{i,1,\alpha} &= -|\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 2) \\ a_{i,2,\alpha} &= \frac{1}{2} |\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 1) \end{aligned} \quad (14)$$

Regarding the neighbor-boundary lattice points, central difference interpolation method is used, thus the interpolation coefficients are given as

$$\begin{aligned} a_{i,0,\alpha} &= \frac{1}{2} \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} - 1) \\ a_{i,1,\alpha} &= 1 - \Delta \xi_{up,i,\alpha}^2 \\ a_{i,2,\alpha} &= \frac{1}{2} \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} + 1) \end{aligned} \quad (15)$$

For the GILBM, the time step should be recalculated to fulfil the stability and accuracy requirements (Breuer, 2000; Karabelas, 2010). In this paper, a local time step δt is applied, and it is determined by

$$\delta t = \lambda \min_{i,\alpha} \left| \frac{1}{\tilde{e}_{i,\alpha} l_\xi} \right| \quad (16)$$

where λ is the CFL number ($\lambda \in (0, 1]$), and it is fixed at $\lambda = 1.0$ in the present work. This strategy guarantees that when a particle streams from position A to position B over a time step, position B will be limited in the neighboring lattice cells of position A. Then, the interpolation step could be implemented successfully. As mentioned in the introduction, the local time step determined by Eq. (16) is generally less than one, thus the dimensionless relaxation time is increased. In this aspect, the numerical stability of this GILBM-based model is improved, which can obtain the similar effect by using central moments or multiple relaxation times as those in the work of Hajabdollahi et al. (Hajabdollahi and Premnath, 2018; Hajabdollahi et al., 2019). Macroscopic parameters, including density ρ , velocity $\mathbf{u}$, and temperature T , are related to f_i and h_i as

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{e}_i, \quad T = \sum_i h_i \quad (17)$$

2.2. Thermal LBM-LES model in body-fitted coordinate system

By introducing the LES into the above GILBM-based thermal LB model, a thermal LBM-LES model in body-fitted coordinate system is obtained. Firstly, a space filtering operation is performed as

$$\bar{\varphi}(\xi) = \int \varphi(\xi') G(\xi, \xi') d\xi' \quad (18)$$

where φ can be density, velocity, temperature, etc., while $\bar{\varphi}$ is the physical quantity after filtering process. G is a spatial filter function, and the standard box filter is used in this paper as

$$G(\xi, \xi') = \begin{cases} 1/\Delta & , |\xi - \xi'| \leq \Delta/2 \\ 0 & , |\xi - \xi'| > \Delta/2 \end{cases} \quad (19)$$

where Δ is the filter width equivalent to the lattice length. In order to apply the idea of LES into the LB model, the density and temperature distribution functions f_i and h_i should be filtered as well. In fact, the obtained evolution equations own the same forms as they are before filtering process. In this LES-based LB model, the equivalent viscosity and diffusivity, ν_{total} and κ_{total} , are defined as

$$\nu_{total} = \nu_0 + \nu_t, \quad \kappa_{total} = \kappa_0 + \kappa_t \quad (20)$$

where, ν_0 and κ_0 represent the physical viscosity and diffusivity, while ν_t and κ_t are the eddy viscosity and diffusivity, respectively. For the flow with specific Reynolds number (Re) and Prandtl number (Pr), ν_0 and κ_0 are determined by

$$\nu_0 = UL/\text{Re}, \quad \kappa_0 = \nu_0/\text{Pr} \quad (21)$$

where U and L are the characteristic velocity and length, respectively. As for the eddy viscosity, it is represented in the Smagorinsky model as

$$\nu_t = (C_s \Delta)^2 |\bar{S}| \quad (22)$$

where C_s is called the Smagorinsky constant, and it is adjustable. $|\bar{S}|$ is the magnitude of strain rate tensor (Chang et al., 2013) calculated by

$$|\bar{S}| = \frac{\sqrt{\tau_f^2 + \frac{18\sqrt{2}Q(C_s\Delta)^2}{\rho}}}{6(C_s\Delta)^2} - \tau_f \quad (23)$$

where, $\tau_f = \nu_0/(c_s^2 \delta t) + 0.5$, Q is the magnitude of the non-equilibrium stress tensor Π_{mn} as

$$Q = \Pi_{mn} \Pi_{mn} \quad (24)$$

$$\Pi_{mn} = \sum_i e_{im} e_{in} (f_i - f_i^{eq}) \quad (25)$$

The eddy diffusivity κ_t is linked with eddy viscosity ν_t by defining the sub-grid Prandtl number Pr_t as

$$Pr_t = \frac{\nu_t}{\kappa_t} \quad (26)$$

Once the equivalent viscosity and diffusivity are obtained by Eqs. (18)–(24), the equivalent relaxation time $\tau_{f, total}$ and $\tau_{T, total}$ can be determined by

$$\nu_{total} = c_s^2 (\tau_{f, total} - 0.5) \delta t \quad (27)$$

$$\kappa_{total} = c_s^2 (\tau_{T, total} - 0.5) \delta t \quad (28)$$

Finally, the thermal LBM-LES model in body-fitted coordinate system is established by replacing the relaxation times in Eqs. (1) and (2) with the equivalent ones in Eqs. (27) and (28). Given the improved numerical stability in simulating turbulent thermal flow on non-uniform grid, this new model has similar advantages to the DUGKS (Wang et al., 2017).

3. Numerical simulation and results discussion

3.1. Simulation set up

In this section, a numerical investigation of the thermal flow around a circular cylinder in a wide range of Reynolds numbers is conducted. Fig. 2 gives the schematic diagram of present issue, in which the fluid flow past a surface-heated circular cylinder with the streaming velocity $U_\infty = 0.1$. Initially, the fluid temperature is kept at a low value ($T_f = 0.0$), yet the cylinder surface temperature holds a constant high one ($T_s = 1.0$). An extremely dense grid in the near-wall region while a relatively coarse one in the far-field is adopted to avoid any kind of wall functions and tune the balance between the numerical accuracy and efficiency (Breuer, 2000; Stiebler et al., 2011). Hence, as shown in Fig. 3, the computational domain is divided by a uniform grid, but the physical domain is covered with a polar coordinate grid and the coordinates of the grid nodes are governed by

$$\begin{cases} r_\eta = \frac{D}{2} e^{a_s \eta / NY}, & \eta = 0, 1, \dots, NY \\ \theta_\xi = 2\pi \xi / NX, & \xi = 0, 1, \dots, NX \\ z_\gamma = L \gamma / NZ, & \gamma = 0, 1, \dots, NZ \end{cases} \quad (29)$$

where NX , NY and NZ are the grid numbers in ξ , η and γ directions, respectively. D and L are the diameter and axial length of the cylinder with $L = \pi D$. a_s is the factor adjusting the local grid refinement. The

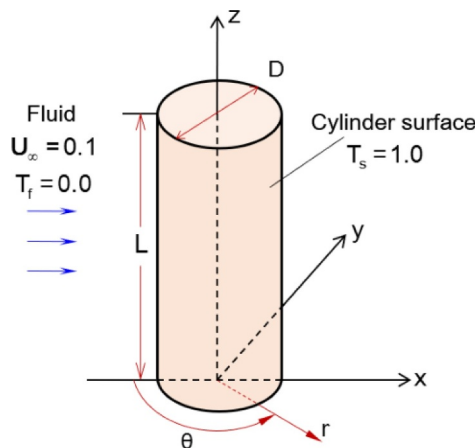


Fig. 2. Schematic diagram for the flow and heat transfer around a circular cylinder.

solid lines AB and CD in Fig. 3(b) represent the grid-cut planes in the x - and z -directions corresponding to the boundaries $A'B'$ and $C'D'$ in the 2D computational plane (Fig. 3(c)).

In order to evaluate the features of fluid flow, the following characteristic parameters: drag coefficient C_D , lift coefficient C_L , Strouhal number St , and pressure coefficient C_p , are calculated as

$$C_D = \frac{2\mathbf{F} \cdot \mathbf{x}}{\rho U_\infty^2 D}, \quad C_L = \frac{2\mathbf{F} \cdot \mathbf{y}}{\rho U_\infty^2 D}, \quad St = f \frac{D}{U_\infty}, \quad C_p = \frac{2(p - p_\infty)}{\rho U_\infty^2} \quad (30)$$

where, f is the vortex shedding frequency that can be obtained by measuring the final period of the lift coefficient in unsteady regime. p_∞ is the pressure at the far-field. $\mathbf{F}$ is the total force acted on the cylinder surface which arises from the surface pressure p and shear stress $\boldsymbol{\tau}$ (He and Doolen, 1997a). In this paper, the total force is determined by integrating the stress tensor around the cylinder surface as $\mathbf{F} = \int \boldsymbol{\sigma} \cdot \mathbf{n} d\mathbf{s}$ (He and Doolen, 1997b). Here, the stress tensor $\boldsymbol{\sigma}$ is calculated by

$$\begin{cases} \sigma_{\alpha\beta} = -p\delta_{\alpha\beta} + \tau_{\alpha\beta} = -p\delta_{\alpha\beta} + 2\rho\nu_0 \varepsilon_{\alpha\beta} \\ \varepsilon_{\alpha\beta} = -\frac{1}{2\rho c_s^2 \gamma \delta t} \sum_i e_{i\alpha} e_{i\beta} [f_i(\xi, t) - f_i^{eq}(\xi, t)] \end{cases} \quad (31)$$

Local Nusselt number $Nu(\theta)$ and average Nusselt number $\bar{Nu}$ are key parameters for examining the heat transfer rate of the heated cylinder surface. Thus, the local Nusselt number around the cylinder surface is defined as

$$Nu(\theta, z) = -\frac{D}{(T_s - T_f)} \left(\frac{\partial T}{\partial n_s} \right) \quad (32)$$

where $\partial T / \partial n_s$ is the temperature gradient on the cylinder surface. Then, the surface-averaged Nusselt number can be obtained by averaging the local values over the cylinder surface as

$$\bar{Nu} = \frac{1}{2\pi L} \int_0^L \int_0^{2\pi} Nu(\theta, z) d\theta dz \quad (33)$$

3.2. Implementation of boundary and initial conditions

Since the governing equations have been transformed from the physical domain into the computational domain, the boundary condition should be transformed as well. As shown in Fig. 4, the boundary relationship is labelled in a spanwise cut of the computational domain. In the current study, the non-equilibrium extrapolation scheme proposed by Guo et al. (2002a,b) is adopted to deal with the velocity and thermal boundary conditions at the cylinder surface ($A'C'$). Consequently, the post-collision particle distribution functions $f_i^*(\xi, 0, \gamma, t)$ and $h_i^*(\xi, 0, \gamma, t)$ at the cylinder surface is expressed as

$$\begin{cases} f_i^*(\xi, 0, \gamma, t) = f_i^{eq}(\rho(\xi, 1, \gamma), \mathbf{u}(\xi, 0, \gamma, t)) \\ \quad + \left(1 - \frac{1}{\tau_f}\right) [f_i(\xi, 1, \gamma, t) - f_i^{eq}(\xi, 1, \gamma, t)] \\ h_i^*(\xi, 0, \gamma, t) = h_i^{eq}(T(\xi, 0, \gamma), \mathbf{u}(\xi, 0, \gamma, t)) \\ \quad + \left(1 - \frac{1}{\tau_T}\right) [h_i(\xi, 1, \gamma, t) - h_i^{eq}(\xi, 1, \gamma, t)] \end{cases} \quad (34)$$

On the other hand, convective boundary condition with average velocity (CBC-AV) (Lou et al., 2013) is considered for the far-field boundary. As for the implementation of the boundary conditions in the spanwise direction and grid-cut plane ($A'B'/C'D'$), the periodic boundary scheme is used.

The numerical simulation starts from an unsymmetric flow field, which is assumed as

$$u_x = U_\infty \frac{Dy}{2r^2}, \quad u_y = -U_\infty \frac{Dx}{2r^2}, \quad u_z = 0.0 \quad (35)$$

With such an unsymmetric flow field setting, the vortex shedding process is triggered when Reynolds number is larger than the critical value ($Re > Re_c \approx 49.0$). If $Re < Re_c$, this asymmetry is gradually

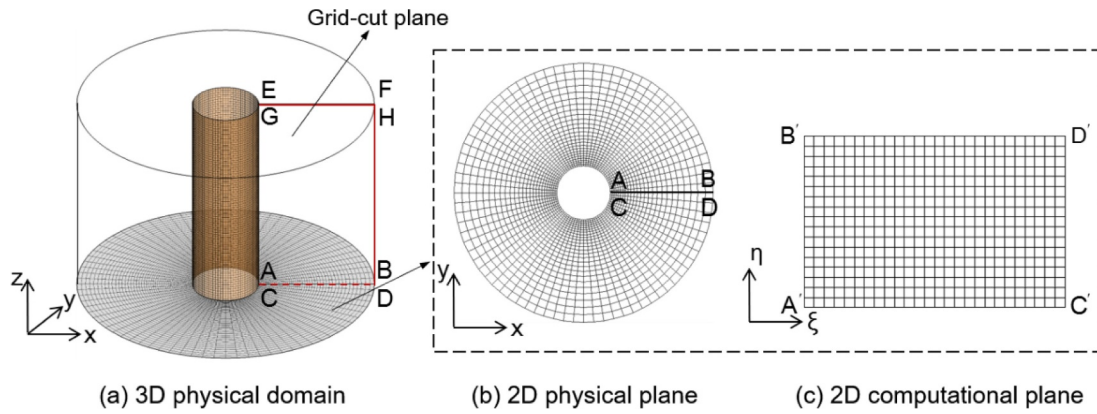


Fig. 3. Grid generation in the (a) 3D physical domain, (b) 2D physical plane and (c) 2D computational plane.

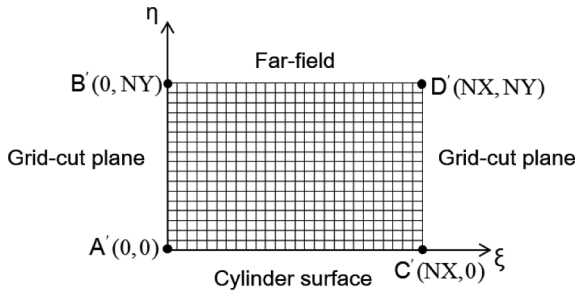


Fig. 4. Boundary correspondence in a spanwise cut of the computational domain.

dissipated by viscosity (He and Doolen, 1997a). The particle distribution functions could generally be initialized with their equilibrium states when the Reynolds number is small. Once the fluid flows at high Reynolds numbers, the non-equilibrium part should be considered at the initialization stage. This can be achieved by following the approach of Mei et al. (2006).

3.3. Results and discussion

In the current study, the flow past a surface-heated circular cylinder is thoroughly investigated over a wide range of Reynolds numbers. As the heat transfer is considered, the influences of Reynolds number and Prandtl number on thermodynamic features of the flow past a circular cylinder are analyzed in detail. Hence, two series of cases categorized by Reynolds number are involved in the numerical simulation. Key parameters used in both cases are all listed in Table 1.

3.3.1. Case A-moderate Reynolds number

For the moderate Reynolds numbers ($Re = 100, 140$, and 150) considered in this part, the flow structure eventually gets into a periodic oscillatory pattern. Since the flow pattern is laminar with no 3D effect under the moderate Reynolds numbers, a 2D simulation with the D2Q9 model in our previous work (Li et al., 2015) is adoptable. As the present 3D model is derived from this 2D one by reconstructing the Jacobian conversion and the interpolation functions, the LES can be introduced into it in the same manner as that in Section 2.2. Preliminarily, several

dynamic parameters for $Re = 100$ and 150 are calculated and compared with previous experimental and numerical results in Table 2. Here, $\bar{C}_D$ is the average drag coefficient, while ΔC_D and ΔC_L are the drag and lift oscillations, respectively. For all the dynamic parameters obtained, agreement between present results and those in previous work is quite satisfactory. In addition, time evolutions of the drag and lift coefficients are shown in Fig. 5. It can be observed that the drag coefficient varies almost twice as fast as the lift coefficient for both Reynolds numbers, which agrees well with the results of He and Doolen (1997a) and Dong et al. (2014). With the increase of Reynolds number, the drag and lift oscillations and vortex shedding frequency (St) experience an enhancement accordingly. The above-mentioned results demonstrate that for the laminar flow, the dissipation introduced by the SGS model is negligible and it has little influence on the numerical results, which is in line with that reported by Chang et al. (2013).

The periodic behavior of the temperature field in a single time period (Te) is discussed as well by taking the effect of Prandtl numbers into account. In Fig. 6, the shedding motion of the isotherm contours is observed periodically for $Re = 100$. As these shed isotherm contours move downstream, a neck connecting its downstream tip with the upstream main body is formed gradually. The neck is then stretched upward or downward by the shed isotherm contours, and finally it is broken to produce isolated isotherm contours. The above-mentioned wave-like behavior repeats during the whole heat transfer process, and it becomes more obvious for $Pr = 10.0$, which is qualitatively consistent with that reported in the previous studies (Patnana et al., 2010; Karniadakis, 1988; Mahfouz and Badr, 2000; Zhang et al., 2008). The heat transfer rate around the cylinder surface can also be evaluated from the isotherm profiles in Fig. 6. It is seen that the isotherm cluster at the front cylinder surface is the densest which indicates the highest heat transfer rate there. When the Prandtl number increases, the isotherm cluster is intensified in the vicinity of the rear cylinder part which leads to an enhancement of the heat transfer. It is proved by the time-averaged values of the surface-averaged Nusselt number ($\bar{Nu}$) listed in Table 3. Meanwhile, the $\bar{Nu}$ obtained by present method is compared with those of Patnana et al. (2010), and the maximum relative error is less than 3.17%.

Table 2

Comparison of dynamic parameters with previous work for $Re = 100$ and 150 .

Re	Work	$\bar{C}_D$	ΔC_D	ΔC_L	St
100	He and Doolen (1997a)	1.287	0.018	0.640	0.161
	Jordan and Fromm (1972)	1.280	0.012	0.540	–
	Present work	1.307	0.017	0.648	0.162
	He and Doolen (1997a)	1.261	0.048	0.980	0.176
150	Dong et al. (2014)	1.297	–	1.041	0.183
	Present work	1.284	0.049	0.999	0.181

Table 1

Parameters used in different cases.

Case	Re	Pr	C_s	Pr_t
A-Moderate Reynolds number	100, 140, and 150	0.71, 1.0, and 10.0	0.12	0.7
B-High Reynolds number	1.4×10^5			

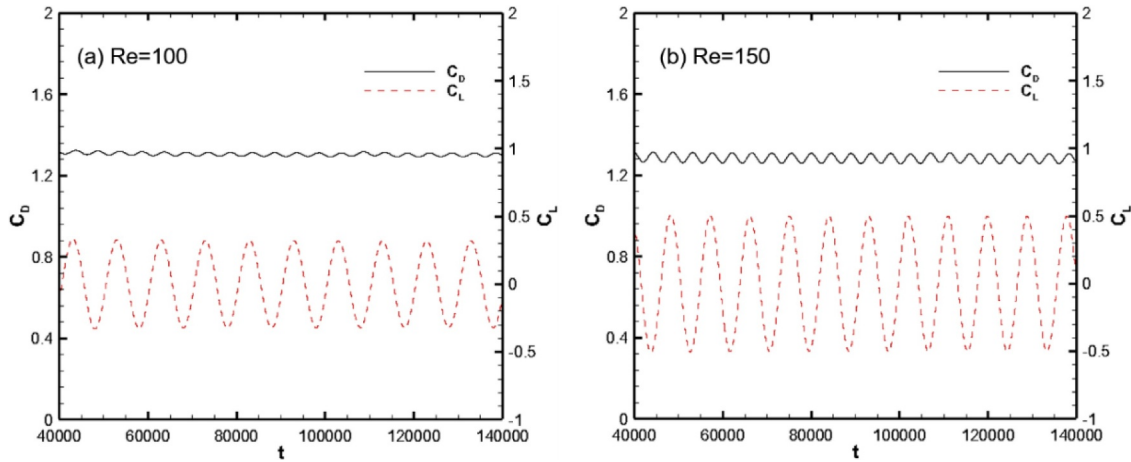


Fig. 5. Time variations of the drag and lift coefficients for (a) $Re=100$ and (b) $Re=150$.

To provide in-depth illustrations of the heat transfer rate affected by the Prandtl number, periodic variation curves of the local Nusselt number for different Prandtl numbers are portrayed in Fig. 7. It can be found out that the Nusselt number at the front stagnation point changes slightly during one time period, yet those in the vicinity of the rear stagnation point experience a rapid variation. This is stemmed from the periodic shedding motion of isotherm contours in the rear of the cylinder, which greatly influences the heat transfer on the rear cylinder surface. Furthermore, distinct fluctuation of the local Nusselt number at different moments within a time period occurs mainly in a specific region of the rear cylinder surface, as shown in Fig. 8. The respective affected regions for $Pr=0.71$, 1.0 , and 10.0 are specified by the angles $\theta_1 = 112^\circ$, $\theta_2 = 102^\circ$, and $\theta_3 = 78^\circ$. Evidently, the range of affected regions shrinks with the increasing value of Prandtl number. One possible reason is that the clustering of isotherms near the rear stagnation point is intensified when the Prandtl number increases.

Table 3

Comparison of $\langle \overline{Nu} \rangle$ with previous work for $Re=100$ and 140 .

Re	Work	$\langle \overline{Nu} \rangle$ $Pr=0.71$	$Pr=1.0$	$Pr=10.0$
100	Patnana et al. (2010)	5.15	5.86	13.94
	Present	5.14	5.84	13.61
	Relative error	0.19%	0.34%	2.37%
140	Patnana et al. (2010)	–	7.03	16.58
	Present	6.22	6.95	16.07
	Relative error	–	1.13%	3.17%

3.3.2. Case B-high Reynolds number

In this section, the thermal flow in the sub-critical flow regime with high Reynolds number ($Re = 1.4 \times 10^5$) is concentrated. For the anisotropic turbulent flow pattern at present Reynolds number, a 3D simulation must be conducted to capture the fluctuations in all

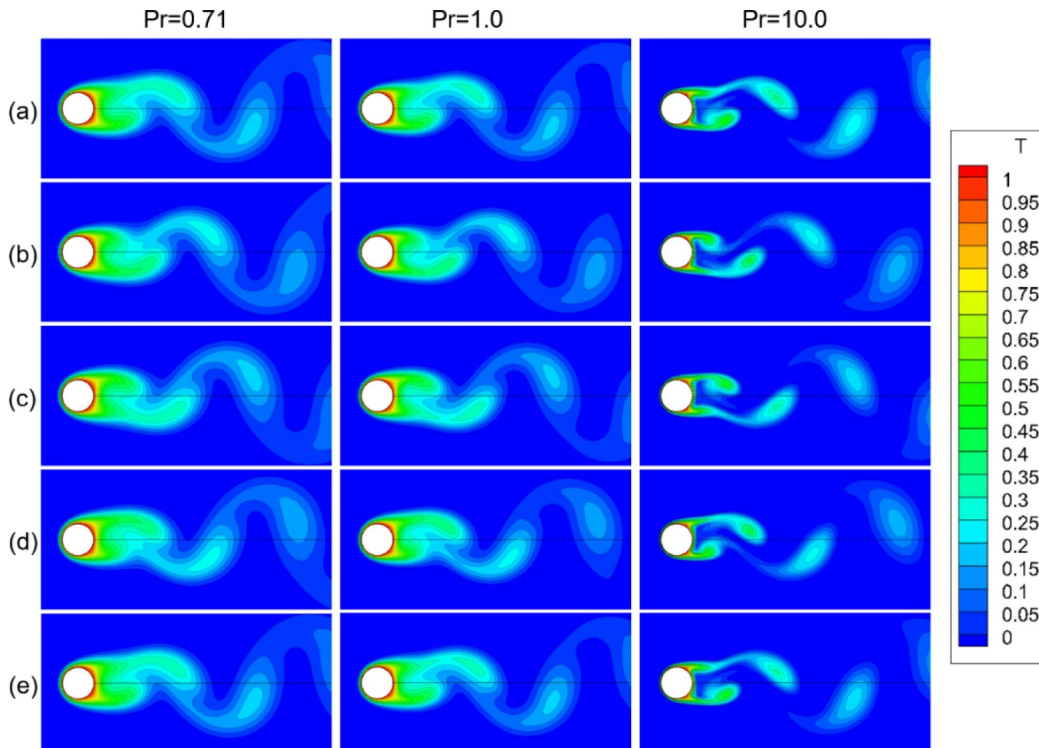


Fig. 6. Time evolutions of the isotherm profile during one time period for different Prandtl numbers: (a) $t=0$, (b) $t=Te/4$, (c) $t=Te/2$, (d) $t=3Te/4$, and (e) $t=Te$.

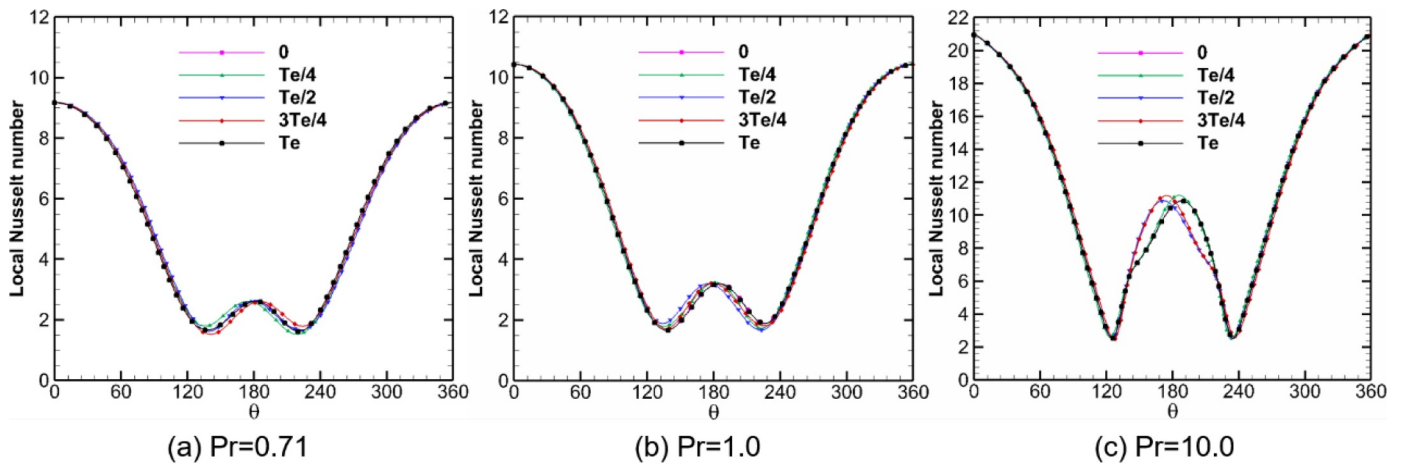


Fig. 7. Time evolutions of the local Nusselt number on cylinder surface during one time period for different Prandtl numbers.

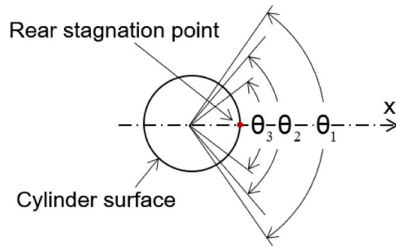


Fig. 8. Fluctuant regions of the local Nusselt number on cylinder surface for different Prandtl numbers.

directions. The grid resolution study is performed in advance, and the results are listed in Table 4 along with a series of data published earlier. It should be emphasized that for the experimentally determined St in Table 4, the values varying around 0.2 gets approval from most people (Breuer, 2000; Achenbach, 1968; Son and Hanratty, 2006; Zdravkovich and Bearman, 1998; Fey et al., 1998), while the extremely low Strouhal number ($St = 0.179$) found by Cantwell and Coles (1983) has large deviation. By examining the predicted values of St and $\bar{C}_D$ comprehensively, it can be concluded that the results generated from denser grids agree better with the general accepted values. Thus, the following studies are implemented based on the grids $340 \times 340 \times 64$ with the local grid refinement factor $a_s = 0.5\pi$. The height of the first cell at the cylinder surface ($\Delta r_{\min}/D = 2.32 \times 10^{-3}$) is approximately 10% less than the value used by Breuer (1998). The resulting non-dimensional distance to the cylinder surface (y^+) is kept at the order of 1.0, which is small enough to resolve the boundary layer on the cylinder surface.

For detailed verifications of the flow field predicted by present method, more quantitative comparisons should be made. First, the time-averaged streamwise velocity $\langle U \rangle$ along the centerline ($y/D = 0$) is considered in Fig. 9. It is obvious that present results are more consistent with the experimental data measured by Cantwell and Coles

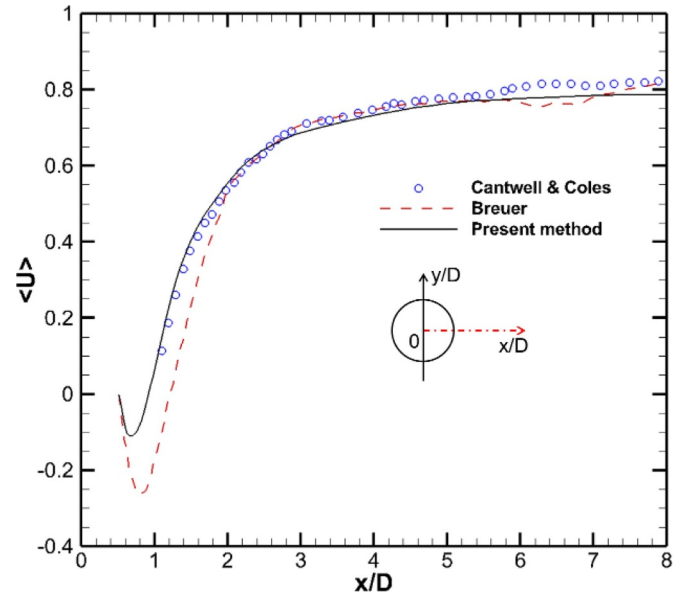


Fig. 9. Comparison of the time-averaged streamwise velocity along the centerline.

(1983) than the numerical results calculated by Breuer (2000), especially in the near wake. Then, both the time-averaged streamwise and vertical velocities ($\langle U \rangle$ and $\langle V \rangle$) along the line ($x/D = 1.0$) are plotted in Fig. 10. Distinct symmetry of the velocity distributions is observed, and the agreement with the published results is satisfactory as well.

Regarding the heat transfer around a circular cylinder, unfortunately, there is a general lack of detailed experimental data at sub-critical Reynolds numbers. In order to make a validation, the time-averaged values of surface-averaged Nusselt numbers ($\bar{Nu}$) predicted by

Table 4
Comparison of dynamic parameters with those in the previous work.

Re	Work	$\bar{C}_D$	St
1.4×10^5	Numerical results of Breuer with coarse grid (Breuer, 2000)	1.218	0.217
	Numerical results of Breuer with fine grid (Breuer, 2000)	1.286	0.203
	Experimental results of Achenbach et al. (Achenbach, 1968; Son and Hanratty, 2006; Zdravkovich and Bearman, 1998; Fey et al., 1998)	1.200	0.200
	Experimental results of Cantwell and Coles (1983)	1.237	0.179
	Present results with grids $160 \times 160 \times 64$	1.234	0.245
	Present results with grids $220 \times 220 \times 64$	1.221	0.231
	Present results with grids $280 \times 280 \times 64$	1.218	0.215
	Present results with grids $340 \times 340 \times 64$	1.216	0.214

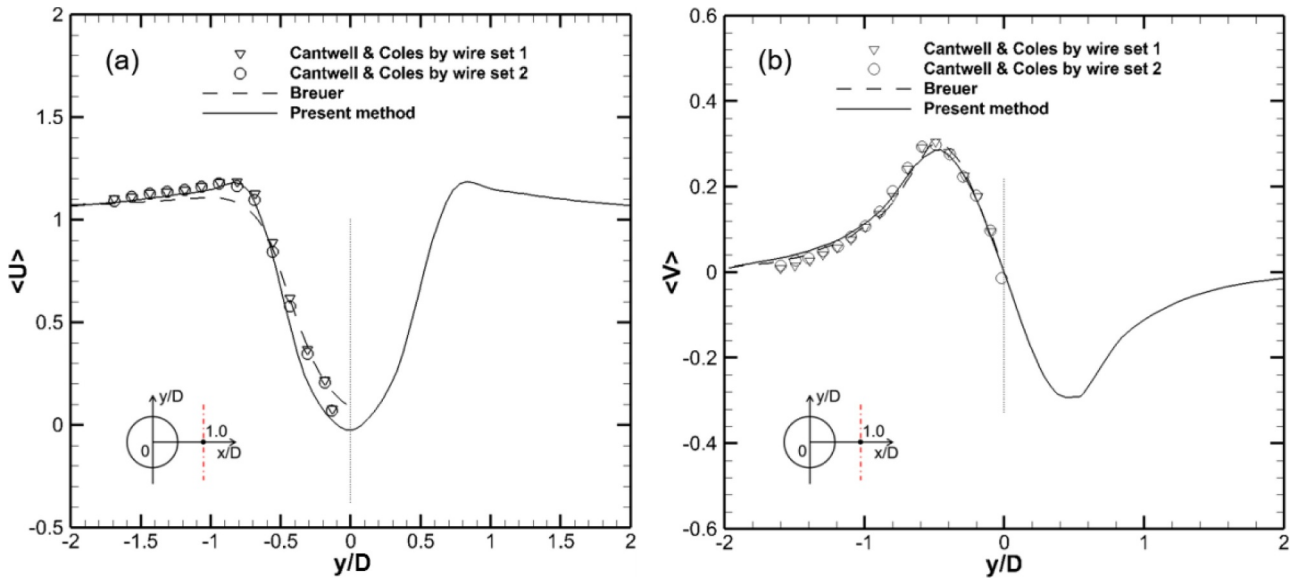


Fig. 10. Comparisons of the time-averaged (a) streamwise velocity and (b) vertical velocity along the line $x/D = 1.0$.

Table 5

Comparison of $\langle \overline{Nu} \rangle$ with those in the previous work for $Re = 1.4 \times 10^5$.

Re	Work	$\langle \overline{Nu} \rangle$		
		Pr = 0.71	Pr = 1.0	Pr = 10.0
1.4×10^5	Churchill and Bernstein (1977)	251.18	287.66	671.46
	Present work	238.39	272.44	638.36
Relative error		5.09%	5.29%	4.93%

present method are compared with those generated from the empirical correlation of Churchill and Bernstein (1977). As shown in Table 5, the numerical results for different Prandtl numbers at $Re = 1.4 \times 10^5$ are all slightly underpredicted. The maximum deviation is 5.29% from the value predicted by the empirical correlation. Considering the turbulent effects and limitations of the empirical correlation, present results can be viewed as acceptable ones.

In terms of the heat transfer in the near-wake region for various Prandtl numbers, it is portrayed by the time-averaged temperature ($\langle T \rangle$) distributions along the centerline $y/D = 0$ and the vertical line $x/D = 1.0$ in Fig. 11. Despite that the Prandtl number changes, the variation tendencies of the time-averaged temperature for different Prandtl numbers maintain a high degree of similarity along both lines. As shown in Fig. 11(a), the time-averaged temperature declines rapidly in a short distance starting from the rear stagnation point. After a short-lived rebound, the downtrend continues along the centerline $y/D = 0$. This changing trend of the time-averaged temperature is caused by the vortex shedding in the near-wake region. For that in Fig. 11(b), the time-averaged temperature distributes symmetrically along the vertical line $x/D = 1.0$, and its peak value appears at the intersection of the two lines. Most notably, the curves of time-averaged temperature for different Prandtl numbers almost overlap, which indicates that the Reynolds number dominates the heat transfer in the near-wake region rather than the Prandtl number at present sub-critical flow regime.

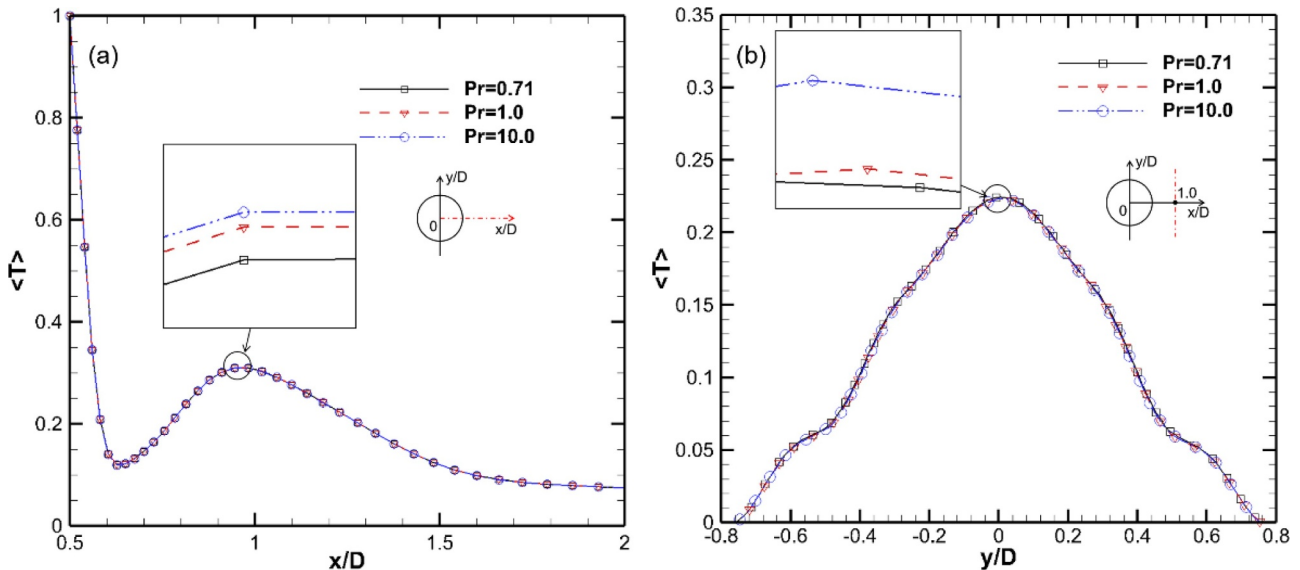


Fig. 11. Time-averaged temperature distributions along lines (a) $y/D = 0$ and (b) $x/D = 1.0$ for various Prandtl numbers.

4. Conclusions

In this paper, by introducing the LES into the GILBM framework, a thermal LBM-LES model in body-fitted coordinate system is established. Given the advantages of GILBM in dealing with complicated geometries and incorporating with local grid refinement techniques, the present model has a great potential to simulate turbulent thermal flow, particularly, in domains with irregular boundaries. For numerical validation, a detailed investigation of the flow and heat transfer around a circular cylinder in a wide range of Reynolds numbers is performed covering both laminar and turbulent flow regimes. Our results are in a general good agreement with the previous experimental and numerical data. In addition, effects of the Reynolds number and Prandtl number on heat transfer are analyzed. It is found out that the heat transfer rate on the cylinder surface is enhanced with the Prandtl number. When the Reynolds number exceeds the critical value, the local Nusselt number fluctuates rapidly in a specific region of the rear cylinder surface affected by the Prandtl number. In the near-wake region, the temperature distribution is significantly influenced by the Prandtl number at moderate Reynolds numbers, while such effects turn to be slight at high Reynolds numbers.

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